

An Optimized Customer Purchase Prediction Framework for E-Commerce Using Weighted Binary Bat Feature Selection and Machine Learning Classifiers

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ABSTRACT.

Machine learning techniques are being utilized to make intelligent decisions as e-commerce platforms continue to generate massive amounts of customer behavioural data. With precise predictions of customer intent before purchase, online retailers can boost engagement, marketing efforts and conversion rates. The study provides an improved method of anticipating customer purchases by utilizing data preprocessing, Weighted Binary Bat Algorithm (WBBA)-based feature selection, and machine learning classification models. To reduce feature dimensionality and improve computational efficiency, the WBBA is used to identify the most relevant customer behavioural attributes. The feature subset generated by the WBBA was used to develop and evaluate SVM and Logistic Regression classifiers for customer purchase prediction. The experimental analysis utilizes an e-commerce customer behaviour dataset, and evaluates the model's performance using accuracy, precision, recall, F1-score, and confusion matrix metrics. This shows that model complexity is reduced and feature optimization significantly improves prediction capability. Logistic Regression achieved an accuracy of 89.1%, whereas the Support Vector Machine classifier outperformed it with an accuracy of 93.4%. In the current e-commerce landscape, this approach can facilitate personalized recommendation systems, targeted advertising, and customer retention tactics. The results demonstrate that a combination of machine learning classifiers and metaheuristic feature selection is an effective means of forecasting customer purchases.

Keywords:

E-Commerce Analytics, Customer Purchase Prediction, Machine Learning, Support Vector Machine, Logistic Regression, Feature Selection, Weighted Binary Bat Algorithm, Predictive Analytics.

I. INTRODUCTION

The rapid advancement of digital commerce has significantly transformed the traditional retail landscape by enabling customers to purchase

products and services through online platforms with greater convenience and accessibility. The widespread adoption of internet services, mobile technologies, and digital payment systems has further accelerated the growth of e-commerce across

the globe [1], [2]. Consequently, online businesses generate large volumes of customer interaction data, including browsing history, clickstream patterns, transaction records, product preferences, and payment activities [1]. Effectively utilizing this data has become a crucial factor in improving business performance and maintaining a competitive advantage in the digital marketplace [2].

Understanding customer purchase behaviour remains one of the major challenges faced by e-commerce organizations. Customers typically interact with multiple products and services before making a final purchase decision, and many users leave online platforms without completing a transaction. Therefore, accurately predicting customer purchase intentions has emerged as an important research area in data analytics and machine learning [1], [2], [3].

Recent developments in artificial intelligence and machine learning have provided powerful tools for extracting meaningful insights from large-scale customer datasets. Predictive models can uncover hidden relationships between customer attributes and purchasing decisions, allowing organizations to implement personalized marketing campaigns, recommendation systems, customer retention strategies, and inventory management solutions [2], [3]. Machine learning algorithms such as Logistic Regression, Decision Trees, Random Forests, Support Vector Machines (SVM), Gradient Boosting, and Neural Networks have demonstrated promising results in customer behaviour prediction and classification tasks [3], [12], [13], [24].

Although numerous machine learning algorithms have been successfully applied to purchase prediction problems, model performance largely depends on the quality and relevance of input features. E-commerce datasets often contain redundant, irrelevant, and noisy attributes that can negatively impact classification accuracy while increasing computational complexity. Feature selection techniques play a critical role in addressing these challenges by identifying the most informative variables and eliminating unnecessary features [10], [11]. Effective feature selection not only improves predictive performance but also enhances model interpretability and reduces training time [10], [11].

In recent years, metaheuristic optimization algorithms have gained considerable attention for feature selection applications. These algorithms mimic natural processes and intelligent behaviours to efficiently explore complex search spaces. Among these approaches, the Bat Algorithm has emerged as an effective optimization technique inspired by the echolocation behaviour of bats [8], [9]. Its binary adaptation is particularly suitable for feature selection problems, where features are either selected or excluded from the final subset [8]. Furthermore, incorporating weighted search mechanisms can improve convergence speed, maintain population diversity, and enhance solution quality during the optimization process [9].

This study proposes a feature-selection-based customer purchase prediction framework for e-commerce environments. The proposed approach integrates a Weighted Binary Bat Algorithm (WBBA) for optimal feature selection with machine learning classifiers, namely Support Vector Machine (SVM) and Logistic Regression (LR). Customer behavioural attributes such as browsing activity, product interactions, transaction values, payment preferences, and website engagement metrics are analysed to predict purchasing intentions. The selected feature subset is then utilized to train and evaluate classification models capable of distinguishing potential buyers from non-buyers more effectively.

The primary objective of this research is to enhance customer purchase prediction performance through intelligent feature optimization. By reducing dataset dimensionality and selecting the most influential attributes, the proposed framework aims to improve predictive accuracy while maintaining computational efficiency. In addition, the study investigates the comparative performance of SVM and Logistic Regression classifiers under optimized feature conditions to identify the most suitable model for customer purchase prediction in e-commerce applications.

II. LITERATURE REVIEW

The rapid growth of e-commerce platforms has attracted significant research attention toward customer behaviour prediction using machine

learning and artificial intelligence techniques [1], [2], [3]. Researchers have proposed various approaches to identify customer purchase intentions, predict conversion rates, and enhance recommendation systems. Customer purchase prediction has become an important application of predictive analytics because it enables organizations to improve marketing effectiveness and customer engagement strategies [1], [2].

Early studies on customer purchase prediction primarily employed statistical methods such as Logistic Regression and Decision Trees to estimate the probability of customer purchases [7]. These approaches offered high interpretability and ease of implementation; however, they often struggled to capture complex nonlinear relationships present in large-scale customer datasets [24].

Support Vector Machine (SVM) has been widely adopted for customer purchase prediction due to its ability to handle high-dimensional data and construct robust classification boundaries [5], [6]. Several studies have reported that SVM-based models outperform traditional statistical approaches when customer interaction data and browsing behaviour are incorporated into the prediction process [6]. Nevertheless, the performance of SVM can be adversely affected by the presence of irrelevant or redundant features, leading to increased computational complexity [10], [11].

Recent research has increasingly focused on ensemble learning techniques such as Random Forest, Gradient Boosting, XGBoost, and AdaBoost [12], [13], [18]. These methods improve predictive performance by combining multiple learning models and reducing classification errors. Although ensemble algorithms often achieve high prediction accuracy, they may require greater computational resources and offer limited interpretability compared to simpler classification models [12], [13].

Deep learning techniques have also gained considerable attention in e-commerce analytics [4], [19]. Models such as Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), and Long Short-Term Memory (LSTM) networks have been applied to analyse customer clickstream

data, transaction histories, and behavioural sequences [4], [19]. These approaches are capable of learning complex behavioural patterns; however, they generally require large training datasets and substantial computational resources [24].

Feature selection remains a critical component of customer purchase prediction systems [10], [11]. E-commerce datasets often contain redundant, noisy, and highly correlated attributes that negatively influence classifier performance [10]. To address this issue, researchers have investigated several feature selection techniques, including Information Gain, Chi-Square Analysis, Recursive Feature Elimination (RFE), Genetic Algorithms (GA), Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), and Bat Algorithm-based approaches [10], [11]. Metaheuristic optimization methods have demonstrated strong potential for identifying optimal feature subsets while reducing dataset dimensionality [10], [11].

Among optimization-based feature selection methods, the Bat Algorithm has emerged as an effective technique inspired by the echolocation behaviour of bats [8], [9]. Binary variants of the algorithm have been successfully applied to feature selection problems in which candidate features are represented using binary decision variables [8]. However, conventional Binary Bat Algorithms may suffer from premature convergence and insufficient exploration in complex search spaces [9]. To overcome these limitations, weighted search strategies have been introduced to improve convergence speed, maintain population diversity, and enhance optimization performance [9].

Despite the significant progress achieved in customer purchase prediction research, several challenges remain unresolved. Many existing studies emphasize classification accuracy while giving limited attention to feature optimization and computational efficiency [10], [11]. The use of large feature sets often increases model complexity and reduces interpretability [10]. In addition, only a limited number of studies have explored the integration of Weighted Binary Bat Optimization with customer purchase prediction models in e-commerce environments [8], [9]. Furthermore, comparative investigations involving optimized

Support Vector Machine and Logistic Regression classifiers remain relatively limited in the existing literature [1], [2], [3].

To address these research gaps, the present study proposes a feature-selection-based customer purchase prediction framework that integrates Weighted Binary Bat Algorithm (WBBA) optimization with machine learning classifiers. The proposed approach aims to improve predictive performance while reducing feature dimensionality and computational overhead. By combining optimization-based feature selection with robust classification techniques, the framework provides a practical and scalable solution for intelligent customer purchase prediction in modern e-commerce environments.

RESEARCH GAP ANALYSIS:

EXISTING LIMITATION	IMPACT	PROPOSED SOLUTION
Large number of irrelevant features	Reduced prediction accuracy	Weighted Binary Bat Feature Selection
High computational complexity	Increased training time	Dimensionality reduction through optimization
Limited feature optimization studies	Poor model efficiency	Intelligent feature subset generation
Lack of comparative analysis	Unclear model suitability	Comparison of SVM and Logistic Regression
Weak personalization capability	Lower customer engagement	Accurate purchase intention prediction

MOTIVATION OF THE STUDY

The increasing demand for personalized shopping experiences has created a need for intelligent systems capable of accurately identifying potential customers before purchase completion [1], [2], [3]. Accurate prediction of customer purchase intentions can help e-commerce organizations improve customer engagement, optimize marketing strategies, and increase conversion rates [1], [2]. At the same time, the presence of redundant and irrelevant features in customer datasets can negatively affect prediction accuracy and computational efficiency [10], [11]. Feature optimization combined with machine learning techniques offers an effective approach for addressing these challenges [8], [10].

Motivated by these factors, this study investigates the effectiveness of Weighted Binary Bat Optimization in improving customer purchase prediction performance while maintaining model simplicity, reducing computational complexity, and enhancing classification accuracy.

III. PROPOSED METHODOLOGY

3.1 OVERVIEW OF THE PROPOSED FRAMEWORK

The proposed framework is designed to predict customer purchase intentions in an e-commerce environment by integrating data preprocessing, feature selection, and machine learning classification techniques. The framework employs a Weighted Binary Bat Algorithm (WBBA) to identify the most relevant customer attributes before model training. The selected feature subset is subsequently used to train Support Vector Machine (SVM) and Logistic Regression (LR) classifiers for customer purchase prediction.

The proposed framework consists of five major stages:

1. Data Collection
2. Data Preprocessing
3. Feature Selection using WBBA
4. Classification using SVM and LR
5. Performance Evaluation

The overall architecture aims to reduce feature dimensionality, eliminate redundant attributes, and improve classification efficiency while maintaining strong predictive performance.

3.2 DATASET DESCRIPTION

The study utilizes a publicly available e-commerce customer behaviour dataset containing customer interaction and transaction-related information. The dataset includes attributes associated with browsing activity, customer engagement, transaction value, and payment preferences. These variables provide valuable insights into customer purchasing behaviour and support the development of predictive models for purchase intention analysis.

The dataset consists of both numerical and categorical attributes. Customer behavioural indicators were selected because they significantly influence online purchasing decisions and can contribute to the accurate prediction of customer purchase intentions.

The primary dataset attributes are presented in Table I.

TABLE I
DATASET ATTRIBUTES

Attribute	Description
Customer ID	Unique customer identifier
Product Category	Product category visited by the customer
Payment Method	Preferred payment option used by the customer
Purchase Value	Total transaction value
Time on Site	Duration spent on the website
Clicks on Site	Number of customer interactions on the platform

Attribute	Description
Purchase Status	Target variable indicating Purchase or No Purchase

Among these attributes, Purchase Status serves as the target variable and represents whether a customer successfully completed a purchase transaction. The remaining attributes are used as predictive features during model development and evaluation.

3.3 DATA PREPROCESSING

Data preprocessing is an essential step in the proposed framework to improve data quality and ensure reliable model training. Raw e-commerce datasets often contain missing values, inconsistent formats, and attributes with varying numerical scales. Therefore, preprocessing techniques are applied before feature selection and classification.

Missing Value Treatment

Missing values are handled using statistical imputation techniques. Numerical attributes are replaced using mean-value substitution to preserve data consistency and minimize information loss.

Equation (1): Mean Imputation

$$X_i = (1/n) \sum X_j$$

where:

X_i = imputed value

n = total number of observations

The imputation process ensures that incomplete records do not negatively affect model performance.

Data Transformation

E-commerce datasets frequently contain categorical attributes such as product categories and payment methods. These attributes are converted into numerical representations using label encoding techniques. This transformation enables machine learning algorithms to process categorical information efficiently.

Feature Normalization

The dataset attributes may have different numerical ranges, which can affect model performance. To prevent feature scale dominance, numerical attributes are normalized using Min-Max scaling.

Equation (2): Min-Max Normalization

$$X_{\text{norm}} = (X - X_{\text{min}}) / (X_{\text{max}} - X_{\text{min}})$$

where:

X = original feature value

X_{min} = minimum feature value

X_{max} = maximum feature value

The normalization process improves model convergence and ensures that all features contribute equally during training.

3.4 WEIGHTED BINARY BAT ALGORITHM (WBBA)

Feature selection plays a critical role in predictive modelling by identifying the most informative attributes and reducing dataset dimensionality [10], [11]. The proposed framework employs a Weighted Binary Bat Algorithm (WBBA) to generate an optimal feature subset before classification.

The Bat Algorithm is a metaheuristic optimization technique inspired by the echolocation behaviour of microbats [8], [9]. In the proposed approach, each bat represents a candidate feature subset, while its movement through the search space is controlled by frequency, velocity, and position parameters.

Frequency Update

The frequency of each bat is updated as follows [8]:

Equation (3): Frequency Update

$$f_i = f_{\text{min}} + (f_{\text{max}} - f_{\text{min}})\beta$$

where:

f_i = frequency of bat i

f_{min} = minimum frequency

f_{max} = maximum frequency

β = random number in the range [0,1]

Velocity Update

The velocity of each bat is updated according to:

Equation (4): Velocity Update

$$v_{it} = v_{it-1} + (x_{it-1} - x_{\text{best}})f_i$$

where:

v_{it} = velocity of bat i at iteration t

x_{it-1} = previous position

x_{best} = global best solution

f_i = frequency parameter

Position Update

The position of each bat is updated using:

Equation (5): Position Update

$$x_{it} = x_{it-1} + v_{it}$$

where:

x_{it} = current position of bat i

v_{it} = current velocity

A binary conversion mechanism transforms continuous positions into binary feature-selection decisions, enabling the algorithm to determine whether a feature should be selected or discarded [8].

Weighted Fitness Function

To simultaneously maximize classification performance and minimize the number of selected features, a weighted fitness function is employed [9].

Equation (6): Fitness Function

$$\text{Fitness} = \alpha(1 - \text{Accuracy}) + \beta(N_s / N_t)$$

where:

Accuracy = classification accuracy

N_s = number of selected features

N_t = total number of available features

α and β = weighting coefficients

The objective of the optimization process is to identify the smallest possible feature subset while maintaining high predictive performance. By eliminating redundant and irrelevant attributes, the WBBA improves classification efficiency, reduces computational complexity, and enhances model interpretability.

3.5 LOGISTIC REGRESSION CLASSIFIER

Logistic Regression (LR) is employed as a baseline classification model due to its simplicity, interpretability, and effectiveness in binary classification problems [7]. The algorithm estimates the probability that a customer will complete a purchase based on the selected behavioural attributes.

Logistic Regression uses the sigmoid function to transform the linear combination of input features into a probability value ranging between 0 and 1.

Equation (7): Sigmoid Function

$$P(Y = 1) = 1 / (1 + e^{-z})$$

where:

$P(Y = 1)$ = probability of customer purchase

e = Euler's constant

z = linear predictor

The linear predictor is expressed as:

Equation (8): Logistic Regression Model

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

where:

β_0 = intercept term

$\beta_1, \beta_2, \dots, \beta_n$ = model coefficients

$x_1, x_2, \dots, x_n$ = input features

The classifier predicts customer purchase behaviour based on a predefined probability threshold. If the predicted probability exceeds the threshold value, the customer is classified as a potential buyer; otherwise, the customer is classified as a non-buyer.

The advantages of Logistic Regression include:

- Simplicity and ease of implementation
- Fast training and prediction
- High interpretability of results
- Suitability for binary classification tasks

Because of these characteristics, Logistic Regression serves as an effective baseline model for evaluating customer purchase prediction performance [7].

3.6 SUPPORT VECTOR MACHINE CLASSIFIER

Support Vector Machine (SVM) is employed as the primary classification model because of its strong generalization capability and effectiveness in handling high-dimensional datasets [5], [6]. SVM is designed to establish the best decision boundary that distinguishes purchasing customers from non-purchasing customers by maximizing the margin between the two classes.

The decision boundary is represented as:

Equation (9): SVM Hyperplane

$$w^T x + b = 0$$

where:

w = weight vector

x = feature vector

b = bias parameter

The objective of SVM is to maximize the separation margin between the two classes.

Equation (10): Margin Maximization

$$\text{Margin} = 2 / \|w\|$$

where:

$\|w\|$ = magnitude of the weight vector

A larger margin generally improves the classifier's ability to generalize to unseen data and reduces the risk of misclassification [6].

To handle complex customer behaviour patterns, SVM can employ various kernel functions that transform data into higher-dimensional feature spaces. Commonly used kernels include:

- Linear Kernel
- Polynomial Kernel
- Radial Basis Function (RBF) Kernel

Among these, the RBF kernel is widely used for modelling nonlinear decision boundaries in customer behaviour prediction tasks [6].

By constructing an optimal hyperplane and maximizing class separation, SVM provides robust classification performance and is particularly effective when combined with optimized feature subsets generated through the WBBA-based feature selection process.

3.7 PERFORMANCE EVALUATION METRICS

The effectiveness of the proposed customer purchase prediction framework is evaluated using widely accepted classification performance metrics. These metrics provide a comprehensive assessment of the predictive capability of the developed models.

Accuracy

Accuracy measures the proportion of correctly classified observations among the total number of observations.

Equation (11): Accuracy

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

where:

TP = True Positive

TN = True Negative

FP = False Positive

FN = False Negative

Precision

Precision evaluates the proportion of correctly predicted positive observations among all predicted positive observations.

Equation (12): Precision

$$\text{Precision} = TP / (TP + FP)$$

A higher precision value indicates a lower false-positive rate and improved prediction reliability.

Recall

Recall measures the ability of the classifier to correctly identify actual positive instances.

Equation (13): Recall

$$\text{Recall} = TP / (TP + FN)$$

A high recall value indicates that the model successfully identifies most customers who are likely to complete a purchase.

F1-Score

The F1-Score provides a balanced evaluation of precision and recall.

Equation (14): F1-Score

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

The F1-Score is particularly useful when evaluating classification models on datasets with class imbalance.

Confusion Matrix

A confusion matrix is employed to analyse classification performance by comparing predicted outcomes with actual outcomes. It provides detailed information regarding correctly and incorrectly

classified observations and assists in understanding model behaviour.

The combination of Accuracy, Precision, Recall, F1-Score, and Confusion Matrix analysis provides a comprehensive evaluation of prediction effectiveness and classification performance.

3.8 PROPOSED WORKFLOW

The workflow of the proposed customer purchase prediction framework is illustrated through a sequence of interconnected stages.

Step 1: Customer Data Collection

Customer interaction and transaction-related data are collected from the e-commerce dataset.

Step 2: Data Cleaning and Preprocessing

Missing values are handled, categorical attributes are transformed into numerical representations, and numerical features are normalized to improve data quality.

Step 3: Feature Selection Using WBBA

The Weighted Binary Bat Algorithm is applied to identify the most relevant customer attributes and eliminate redundant features.

Step 4: Model Training

The optimized feature subset is utilized to train Logistic Regression and Support Vector Machine classifiers.

Step 5: Performance Evaluation

The trained models are evaluated using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix analysis.

Step 6: Customer Purchase Prediction

The best-performing model is employed to predict whether a customer is likely to complete a purchase transaction.

The overall workflow can be summarized as follows:

Customer Data Collection

↓

Data Cleaning and Preprocessing

↓

Feature Normalization

↓

WBBA-Based Feature Selection

↓

Logistic Regression Training

↓

Support Vector Machine Training

↓

Performance Evaluation

↓

Customer Purchase Prediction

The proposed methodology enables the accurate identification of potential buyers and supports intelligent decision-making in modern e-commerce environments. By integrating optimization-based feature selection with machine learning classifiers, the framework improves predictive performance while reducing computational complexity.

IV. EXPERIMENTAL SETUP

4.1 EXPERIMENTAL ENVIRONMENT

The proposed customer purchase prediction framework was implemented using Python 3.8 within the Anaconda development environment. Machine learning models were developed using the Scikit-Learn library [14], while NumPy [17] and Pandas [15] were utilized for data preprocessing and manipulation. Data visualization and performance analysis were performed using Matplotlib [16]. The experiments were conducted on a Windows-based computing platform with sufficient computational resources to support model training and evaluation.

For model training and evaluation, the dataset was partitioned into training and testing subsets, with 80% of the samples used for training and the remaining 20% reserved for testing.. The training dataset was used to develop the predictive models, whereas the testing dataset was utilized to evaluate their generalization capability. To ensure reproducibility and consistency, a fixed random seed was applied during data partitioning.

TABLE II
EXPERIMENTAL CONFIGURATION

Parameter	Value
Programming Language	Python 3.8
Development Tool	Anaconda
Machine Learning Library	Scikit-Learn
Data Processing Libraries	Pandas, NumPy
Visualization Tool	Matplotlib
Training Split	80%
Testing Split	20%
Classifiers	SVM, Logistic Regression
Feature Selection Method	Weighted Binary Bat Algorithm

4.2 DATASET DESCRIPTION

The study utilizes an e-commerce customer behaviour dataset containing customer interaction and transaction-related information. The dataset includes variables associated with browsing activities, payment preferences, transaction values, product categories, and website engagement metrics.

The target variable indicates whether a customer successfully completed a purchase transaction. Customer behavioural attributes were selected because they provide valuable insights into purchasing intentions and online shopping patterns.

TABLE III
DATASET ATTRIBUTES

Feature	Description
Customer ID	Unique identifier of customer
Product Category	Category of product viewed
Payment Method	Customer payment preference
Purchase Value	Total transaction value
Time on Site	Duration spent on website
Clicks on Site	Number of customer interactions
Purchase Status	Purchase or Non-Purchase

The dataset contains both numerical and categorical attributes that require preprocessing before model training and evaluation.

4.3 DATA PREPROCESSING AND TRAINING CONFIGURATION

Data preprocessing was performed to improve model reliability and prediction accuracy. Missing values were handled using mean-based imputation techniques to preserve dataset integrity. Categorical variables were transformed into numerical representations using label encoding methods.

Feature scaling was carried out using Min-Max normalization to ensure that all attributes contributed equally during model training. This process prevented features with larger numerical ranges from dominating the learning process.

The Weighted Binary Bat Algorithm (WBBA) [8], [9] was subsequently applied to identify the most informative features. Feature optimization reduced dataset dimensionality and improved computational efficiency [10], [11]. The optimized feature subset was then utilized to train Logistic Regression [7] and Support Vector Machine classifiers [5], [6].

Hyperparameter tuning was performed to obtain suitable model configurations. The SVM classifier employed a linear kernel with optimized regularization parameters, while Logistic Regression utilized an increased iteration limit to ensure model convergence.

4.4 EVALUATION METRICS

The effectiveness of the proposed framework was evaluated using widely accepted classification metrics.

Accuracy

Accuracy measures the proportion of correctly classified observations among the total number of observations.

Equation (15)

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Precision

Precision evaluates the proportion of correctly predicted positive observations among all predicted positive observations.

Equation (16)

$$\text{Precision} = TP / (TP + FP)$$

Recall

Recall measures the ability of the classifier to correctly identify actual positive instances.

Equation (17)

$$\text{Recall} = TP / (TP + FN)$$

F1-Score

The F1-score represents the harmonic mean of Precision and Recall, thereby providing a balanced measure of classification performance.

Equation (18)

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

where:

TP = True Positive

TN = True Negative

FP = False Positive

FN = False Negative

These evaluation metrics provide a comprehensive assessment of classification performance and prediction effectiveness.

V. RESULTS AND DISCUSSION

5.1 FEATURE SELECTION RESULTS

The Weighted Binary Bat Algorithm (WBBA) was employed to identify the most influential features contributing to customer purchase prediction. The optimization process successfully reduced dataset dimensionality while preserving critical behavioural information. The selected features included Purchase Value, Time on Site, Number of Clicks, Product Category, and Payment Method. These attributes exhibited a strong influence on customer purchasing decisions and contributed significantly to prediction performance.

The feature optimization process improved training efficiency by reducing the number of input variables. In addition, the elimination of redundant and less informative attributes enhanced model interpretability and reduced computational complexity [10], [11].

TABLE IV SELECTED FEATURES AFTER WBBA OPTIMIZATION

Selected Feature	Description
Purchase Value	Customer transaction value
Time on Site	Website engagement duration
Clicks on Site	Customer interaction frequency
Product Category	Product preference information
Payment Method	Customer payment preference

5.2 LOGISTIC REGRESSION RESULTS

The Logistic Regression classifier was trained using the optimized feature subset obtained from the feature selection stage. The model demonstrated effective performance in distinguishing purchasing customers from non-purchasing customers.

TABLE V
LOGISTIC REGRESSION PERFORMANCE

Metric	Value (%)
Accuracy	89.10
Precision	87.60
Recall	88.40
F1-Score	88.00

The obtained results indicate that Logistic Regression provides reliable predictive performance while maintaining a high degree of interpretability. However, its capability to model complex nonlinear customer behaviour patterns remains relatively limited [7].

5.3 SUPPORT VECTOR MACHINE RESULTS

The Support Vector Machine classifier was trained using the same optimized feature subset selected through the WBBA process. The classifier demonstrated superior performance compared with Logistic Regression because of its ability to construct optimal decision boundaries and effectively separate customer classes [6].

TABLE VI
SUPPORT VECTOR MACHINE PERFORMANCE

Metric	Value (%)
Accuracy	93.40
Precision	92.70
Recall	93.10
F1-Score	92.90

The results indicate that SVM achieved higher classification accuracy and lower misclassification rates, making it more suitable for customer purchase prediction tasks.

5.4 COMPARATIVE PERFORMANCE ANALYSIS

A comparative evaluation was conducted to analyse the effectiveness of both classifiers.

TABLE VII
COMPARISON OF CLASSIFIER PERFORMANCE

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Logistic Regression	89.10	87.60	88.40	88.00

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Support Vector Machine	93.40	92.70	93.10	92.90

The comparative results demonstrate that the SVM classifier outperformed Logistic Regression across all evaluation metrics. The optimized feature subset generated through WBBA contributed to improved classification performance by selecting the most relevant customer behavioural attributes [8], [9], [10].

5.5 DISCUSSION

The experimental findings demonstrate the effectiveness of integrating feature selection and machine learning techniques for customer purchase prediction. The Weighted Binary Bat Algorithm successfully identified significant behavioural attributes that strongly influence customer purchasing decisions.

Among the evaluated classifiers, Support Vector Machine achieved the highest performance across all evaluation metrics. The classifier effectively handled complex decision boundaries and produced superior Accuracy, Precision, Recall, and F1-Score values. Logistic Regression also generated competitive results but exhibited comparatively lower performance when modelling nonlinear relationships within customer interaction data [6], [7].

The analysis further revealed that customer engagement indicators such as browsing duration, transaction value, and website interaction frequency are strong predictors of purchase intention. Previous studies have similarly reported that behavioural attributes play a significant role in predicting customer purchasing decisions [1], [2], [3].

The findings indicate that optimization-based feature selection can enhance classification performance by reducing feature redundancy and improving computational efficiency [10], [11]. These insights can assist e-commerce organizations in developing personalized recommendation systems, targeted marketing campaigns, and customer retention strategies.

Overall, the proposed WBBA-SVM framework demonstrates strong potential for intelligent customer purchase prediction and data-driven decision support in modern e-commerce environments.

VI. CONCLUSION AND FUTURE WORK

This study presented a feature-selection-based customer purchase prediction framework for e-commerce applications using machine learning techniques. The proposed framework integrated a Weighted Binary Bat Algorithm (WBBA) with Logistic Regression and Support Vector Machine classifiers to improve prediction performance. Data preprocessing, feature optimization, and classification stages were systematically combined to develop an efficient predictive analytics framework capable of identifying customers who are likely to complete a purchase transaction.

The experimental results demonstrated that feature optimization improved model performance by eliminating redundant attributes and reducing computational complexity [10], [11]. The optimized feature subset enabled the classification models to focus on the most relevant customer behavioural characteristics, thereby improving prediction effectiveness. Comparative analysis revealed that the Support Vector Machine classifier achieved superior performance compared with Logistic Regression across all evaluation metrics. This outcome highlights the effectiveness of SVM in handling complex classification boundaries and customer behavioural patterns [6].

The findings indicate that optimized machine learning models can effectively support customer behaviour analysis and business decision-making

processes. The proposed framework provides practical benefits for e-commerce organizations by enabling enhanced customer targeting, personalized recommendation generation, improved marketing effectiveness, and increased conversion rates. Accurate prediction of customer purchase intentions can assist businesses in allocating resources more efficiently and improving overall customer satisfaction.

Future research may focus on integrating advanced deep learning models such as Long Short-Term Memory (LSTM) networks, Artificial Neural Networks (ANN), and Transformer-based architectures to further improve prediction performance [4], [19]. In addition, real-time customer behaviour analytics, large-scale datasets, and hybrid optimization algorithms may be explored to enhance scalability, adaptability, and predictive capability in dynamic e-commerce environments.

The overall results confirm that the integration of optimization-based feature selection and machine learning classification techniques provides an effective and scalable solution for intelligent customer purchase prediction in modern e-commerce systems.

REFERENCES:

- [1] Hendriksen, M., Kuiper, E., Nauts, P., Schelter, S., and de Rijke, M.: 'Analyzing and predicting purchase intent in e-commerce: Anonymous vs. identified customers', arXiv preprint arXiv:2012.08777, 2020.
- [2] De, T.S., Singh, P., and Patel, A.: 'A machine learning and empirical Bayesian approach for predictive buying in B2B e-commerce', arXiv preprint arXiv:2403.07843, 2024.
- [3] Jain, A.M.: 'Predicting e-commerce purchase behavior using a DQN-inspired deep learning model for enhanced adaptability', arXiv preprint arXiv:2506.17543, 2025.
- [4] Equihua, J.P., Nordmark, H., Ali, M., and Lausen, B.: 'Modelling customer churn for the retail industry in a deep learning based sequential framework', arXiv preprint arXiv:2304.00575, 2023.
- [5] Vapnik, V.: 'The Nature of Statistical Learning Theory' (Springer, New York, USA, 2013).
- [6] Cortes, C., and Vapnik, V.: 'Support-vector networks', Machine Learning, 1995, 20, (3), pp. 273–297.
- [7] Hosmer, D.W., Lemeshow, S., and Sturdivant, R.X.: 'Applied Logistic Regression' (Wiley, USA, 2013, 3rd edn.).
- [8] Yang, X.S.: 'A new metaheuristic bat-inspired algorithm'. Proc. Nature Inspired Cooperative Strategies for Optimization, Springer, Berlin, Germany, 2010, pp. 65–74.
- [9] Yang, X.S.: 'Bat algorithm for multi-objective optimisation', International Journal of Bio-Inspired Computation, 2011, 3, (5), pp. 267–274.
- [10] Chandrashekar, G., and Sahin, F.: 'A survey on feature selection methods', Computers and Electrical Engineering, 2014, 40, (1), pp. 16–28.
- [11] Guyon, I., and Elisseeff, A.: 'An introduction to variable and feature selection', Journal of Machine Learning Research, 2003, 3, pp. 1157–1182.
- [12] Breiman, L.: 'Random forests', Machine Learning, 2001, 45, (1), pp. 5–32.
- [13] Friedman, J.H.: 'Greedy function approximation: A gradient boosting machine', Annals of Statistics, 2001, 29, (5), pp. 1189–1232.
- [14] Pedregosa, F., Varoquaux, G., Gramfort, A., et al.: 'Scikit-learn: Machine learning in Python', Journal of Machine Learning Research, 2011, 12, pp. 2825–2830.

- [15] McKinney, W.: 'Data structures for statistical computing in Python'. Proc. Python in Science Conference, Austin, USA, 2010, pp. 56–61.
- [16] Hunter, J.D.: 'Matplotlib: A 2D graphics environment', Computing in Science and Engineering, 2007, 9, (3), pp. 90–95.
- [17] Harris, C.R., Millman, K.J., van der Walt, S.J., et al.: 'Array programming with NumPy', Nature, 2020, 585, pp. 357–362.
- [18] Chen, T., and Guestrin, C.: 'XGBoost: A scalable tree boosting system'. Proc. ACM SIGKDD, San Francisco, USA, 2016, pp. 785–794.
- [19] Goodfellow, I., Bengio, Y., and Courville, A.: 'Deep Learning' (MIT Press, Cambridge, USA, 2016).
- [20] Géron, A.: 'Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow' (O'Reilly Media, USA, 2022, 3rd edn.).
- [21] Han, J., Kamber, M., and Pei, J.: 'Data Mining: Concepts and Techniques' (Morgan Kaufmann, USA, 2011).
- [22] Aggarwal, C.C.: 'Data Mining: The Textbook' (Springer, Switzerland, 2015).
- [23] Tan, P.N., Steinbach, M., and Kumar, V.: 'Introduction to Data Mining' (Pearson, USA, 2018, 2nd edn.).
- [24] Bishop, C.M.: 'Pattern Recognition and Machine Learning' (Springer, New York, USA, 2006).
- [25] Murphy, K.P.: 'Machine Learning: A Probabilistic Perspective' (MIT Press, Cambridge, USA, 2012).